

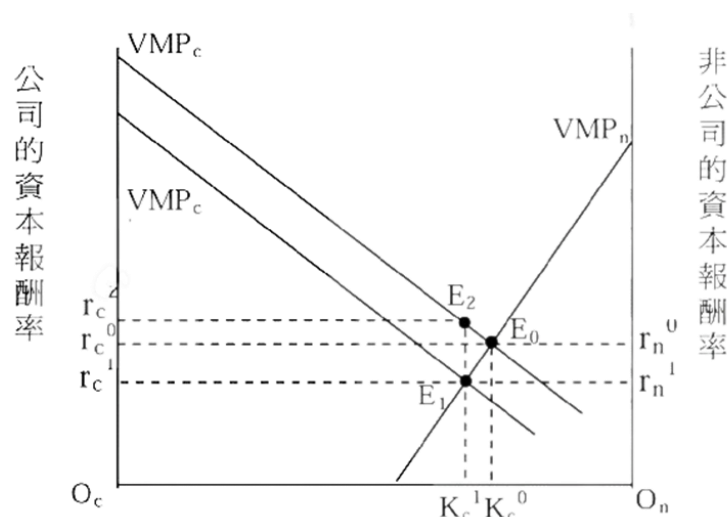
財政學 HW2

1.(112考古) Assume there are two sectors in the economic system, the corporate sector and the non-corporate sector. The marginal value product of capital in the corporate sector is $VMP_c = 150 - K_c$, that K_c is the amount of capital used in the corporate sector. The marginal value product of capital in the non-corporate sector is $VMP_n = 130 - 3K_n$, that K_n is the amount of capital used in the non-corporate sector. The total amount of capital in the society is 100 units. Explain with a diagram, in the case without taxation, what are the amounts of capital in the corporate and non-corporate sectors respectively? Furthermore, assume that the government imposes a tax of 12 units per unit of capital on the corporate sector. What are the amounts of capital and the excess burden of taxation in each sector after tax? Please analyze it with graphs, too.

ANS:

(一) 在無課稅的情況下，公司與非公司部門的均衡：

1. 圖形：



(1) 設縱軸分別為公司部門及非公司部門之資本報酬率，分別以 r_c 、 r_n 表示之；橫軸為資本量。

(2) 稅前橫軸為 VMP_c 及 VMP_n 的交會點，及 E_0 點。公司部門的資本報酬率為 r_{c0} ，資本量為 $O_c K_{c0}$ ；非公司部門的資本報酬率為 r_{n0} ，資本量為 $O_n K_{c0}$ 。

2. 解聯立：

$$150 - K_c = 130 - 3K_n$$

$$K_c + K_n = 100$$

$$\text{得：} K_c = 80, K_n = 20$$

分別代入邊際產值，得： $VMP_c = VMP_n = 70$

(二) 每單位課徵12元，稅後公司部門的邊際產出價值，下降為 VMP_c' ； $VMP_c' =$

$$150 - K_c = 138 - 3K_c$$

解聯立：

$$138 - K_c = 130 - 3K_n$$

$$K_c + K_n = 100$$

得： $K_c = 77, K_n = 23$

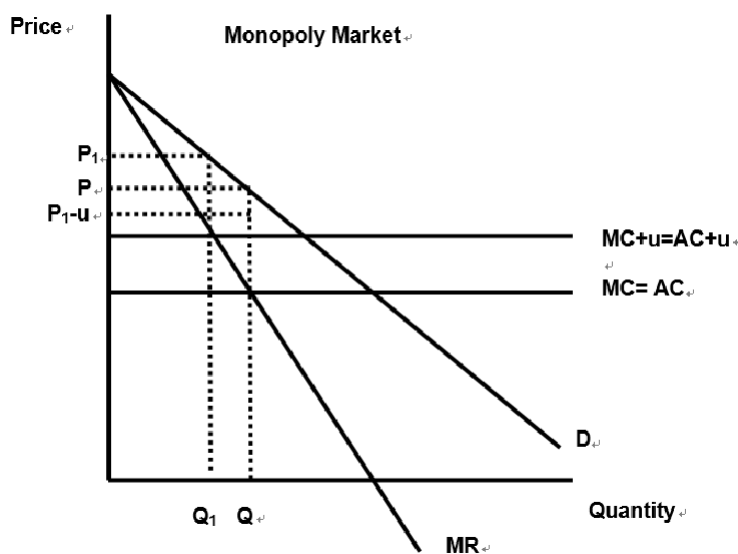
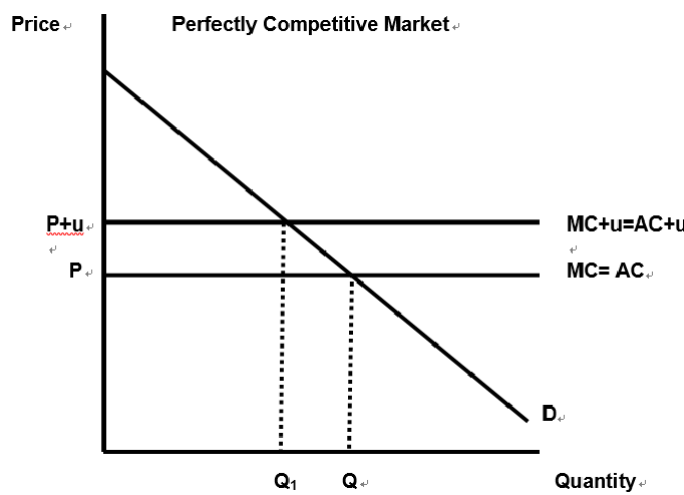
分別代入邊計產值，得： $VMP_c = VMP_n = 61$

租稅的超額負擔為 $\Delta E_0 E_1 E_2 = 1/2 \times 12 \times (80 - 77) = 18$

2.(Rosen 14.3) For commodity X , average cost is equal to marginal cost at every level of output. Assuming that the market for X is competitive and the demand curve is linear, analyze the effects when a unit tax of u dollars is imposed. Now analyze the effects of the same tax assuming that the market for X is a monopoly. Discuss the differences.

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In the competitive case, the full tax is paid by the consumer because the supply curve, given by MC, is perfectly elastic. With a monopoly, the tax is shared between the consumer and the producer depending on the elasticity of demand. (If the tax is imposed on the consumer, the demand curve falls by $\$u$, and the result is the same in both cases).



3.(Rosen 14.5) In an effort to reduce alcohol consumption, the government is considering a \$1 tax on each gallon of liquor sold (the tax is levied on producers). Suppose that the demand curve is $Q^D = 500,000 - 20,000P$ (where Q^D is the number of gallons of liquor demanded and P is the price per gallon), and the supply curve for liquor is $Q^S = 30,000P$ (where Q^S is the number of gallons supplied).

- Compute how the tax affects the price paid by consumers and the price received by producers.
- How much revenue does the tax raise for the government? How much of the revenue comes from consumers, and how much from producers?
- Suppose that the demand for liquor is more elastic for younger drinkers than for older drinkers. Will the liquor tax be more, less, or equally effective at reducing liquor consumption among young drinkers? Explain.

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- a. Before-tax equilibrium: $P = \$10$ and $Q = 300,000$

$$Q_d = Q_s$$

$$500,000 - 20,000P = 30,000P$$

$$500,000 = 50,000P$$

$$P = 10$$

$$Q = 30,000 * 10 = 300,000$$

After-tax equilibrium: $P = \$10.60$ and $Q = 288,000$, producers receive $\$9.60$

$$500,000 - 20,000P = 30,000(P - 1)$$

$$530,000 = 50,000P$$

$$P = 10.6$$

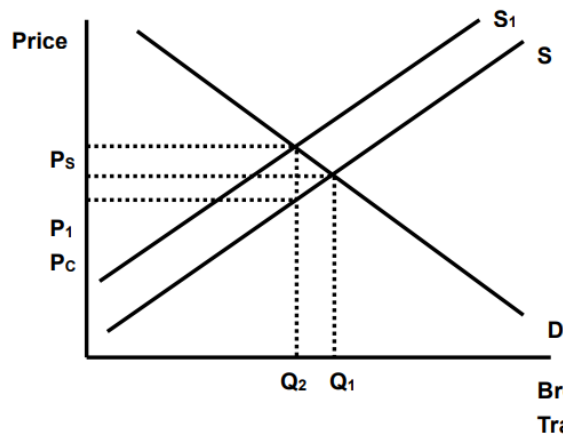
$$Q = 30,000 * (10.6 - 1) = 288,000$$

- b. Revenue = $\$288,000$. Consumers bear 60 percent of the tax burden and producers bear 40 percent. So, $\$172,800$ comes from consumers and $\$115,200$ from producers.
- c. With a more elastic demand curve, quantity consumed will decrease even more as a result of the tax, so the liquor tax will be more effective at reducing consumption among young drinkers.

4.(Rosen14.1) In 2009, it was proposed that the state of Nevada create an entertainment tax that “would require the state’s 25 legal brothels to give the state some money on a per-transaction basis” [Friess, 2009]. Discuss the likely incidence of such a tax. Use an appropriate diagram as the basis for your discussion.

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The market initially is at P_1 Q_1 . Adding a per-customer tax on strip clubs would be a tax on a commodity. Because the producer would be paying the tax, the effective supply curve for suppliers would be S_1 . The producers are happy and the market clears only when PS is paid by the customers for quantity Q_2 . Of this price, $PS - PC$ is the tax paid, where customers pay $PS - P_1$ of the tax and the suppliers pay $P_1 - PC$. The incidence depends on the relative elasticities of supply and demand. Here, the curves are drawn such that the consumers and suppliers face roughly the same incidence. However, if one believed that the supply was relatively more inelastic, then the strip clubs would face a higher incidence.



5.請從《家庭收支調查報告》的統計數據 (如吉尼係數、所得五等分位差距倍數)，說明我國近年的所得分配趨勢。

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請詳閱《家庭收支調查報告》p.7-p.10

https://ebook.dgbas.gov.tw/News_Content_Ebook.aspx?n=3786&s=233664

